

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 1
Date of Assignment: 04-08-2026	Date/Time of Submission:04-08-2026

Part B - Program

a. Develop a PWA that works in offline mode using Service Worker.

Code:

Manifest.json

```
{
  "name": "College Information Portal",
  "short_name": "CIP",
  "short_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

Service Worker

```
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.add("/offline.html")
    })
  );
});
```

```
});
self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch((
      caches.match("/offline.html"))
    )
  );
});
```

Layout.cshtml

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - Practical_7_a</title>
  <script type="importmap"></script>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/Practical_7_a.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/Manifest.json"/>
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home" asp-
action="Index">Practical_7_a</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target=".navbar-collapse" aria-controls="navbarSupportedContent"
          aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
          <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Index">Home</a>
            </li>
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
            </li>
          </ul>
```

```

        </div>
    </div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - Practical_7_a - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
    </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
    if('serviceWorker' in navigator){
        navigator.serviceWorker.register('/service.worker.js');
    }
</script>
</body>
</html>

```

offline.html

```

<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <title>College Information Portal</title>
</head>
<body>
    <h2>Select Your Course</h2>

    <form>
        <label for="course">Choose a course:</label>

        <select id="course" name="course">
            <option value="">-- Select Course --</option>
            <option value="bca">BCA</option>
            <option value="bscit">B.Sc IT</option>

```

```
<option value="btech">B.Tech</option>
<option value="mca">MCA</option>
</select>
```

```
<br><br>
```

```
<input type="submit" value="Submit">
</form>
```

```
</body>
```

```
</html>
```

Output:

The screenshot shows the Chrome DevTools interface. The top panel displays the 'Manifest' editor for 'manifest.json'. The left sidebar shows the 'Application' tab with various categories like Storage, Background services, and Forms. The main content area shows the 'Manifest' editor with sections for Errors and warnings, Identity, Presentation, Protocol Handlers, Icons, and Window Controls Overlay. The 'Service workers' panel is open at the bottom, showing the 'https://localhost:7225/' source, the file 'service-worker.js', and its status as '#461 activated and is running'. The panel includes controls for 'Push', 'Sync', and 'Periodic sync'.

Service workers

☒ Offline ☐ Update on reload ☐ Bypass for network

https://localhost:7225/

Source: [service-worker.js](#)
Received 8/4/2026, 9:04:21 AM

Status: ● #461 activated and is running Stop

Push: Push

Sync: Sync

Periodic sync: Periodic sync

DevTools - localhost:7225/

Elements Console Sources **Network** Performance Memory Application Security Lighthouse Recorder

Preserve log Disable cache Offline

Filter Invert More filters All Fetch/XHR Doc CS

Name	Status	Type	Initiator
localhost	200	document	Other
localhost	(failed) net::ERR_INTER...	fetch	service-worker.js:10
favicon.ico	200	text/html	Other
favicon.ico	(failed) net::ERR_INTER...	fetch	service-worker.js:10



College Information Portal

College Name: MVLU College

Choose a course: -- Select Course -- ▾



College Information Portal

College Name: MVLU College

Choose a course: B.Sc IT ▾

b. Create a Student Record Management PWA.**Code:****Manifest.json**

```
{
  "name": "",
  "short_name": "",
  "start_url": "",
  "display": "",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js

```
const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
```

```

        return response || fetch(e.request);
    })
    );
});

```

_Layout.cshtml

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - Practical_6A_B</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/Practical_6A_B.styles.css" asp-append-version="true" />
    <link rel="manifest" href="Manifest.json"/>
</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home" asp-
action="Index">Practical_6A_B</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target=".navbar-collapse" aria-controls="navbarSupportedContent"
                    aria-expanded="false" aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>

```

```

</header>
<div class="container">
  <main role="main" class="pb-3">
    @RenderBody()
  </main>
</div>

<footer class="border-top footer text-muted">
  <div class="container">
    &copy; 2026 - Practical_6A_B - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
  </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
  if('serviceWorker' in navigator)
  {
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>
</body>
</html>

```

Student.cs

```

namespace Practical_6A_B.Models
{
  public class Students
  {
    public class Student
    {
      public int RollNo { get; set; }
      public string Name { get; set; }
    }
  }
}

```

HomeController.cs

```

using System.Diagnostics;

```

```
using Microsoft.AspNetCore.Mvc;  
using Practical_6A_B.Models;  
using static Practical_6A_B.Models.Students;
```

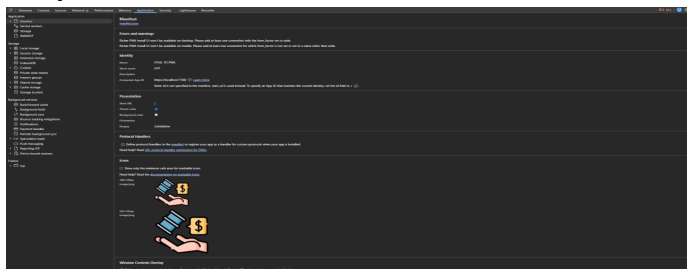
```
public class HomeController : Controller  
{  
    public IActionResult Index()  
    {  
        Student S = new Student();  
  
        S.RollNo = 101;  
        S.Name = "Rahul";  
  
        return View(S);  
    }  
}
```

Index.cshtml

```
@{  
    ViewData["Title"] = "Home Page";  
}
```

```
<div class="text-center">  
    <h1 class="display-4">Welcome</h1>  
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with  
ASP.NET Core</a>.</p>  
    <h1>Offline Page</h1>  
    <h2>Student Record</h2>  
  
    <p>Roll No : @Model.RollNo</p>  
  
    <p>Name : @Model.Name</p>  
</div>
```

Output:



Service workers

☒ Offline ☐ Update on reload ☐ Bypass for network

https://localhost:7225/

Source	service-worker.js Received 8/4/2026, 9:04:21 AM		
Status	● #461 activated and is running <button>Stop</button>		
Push	Test push message from DevTools.	<button>Push</button>	
Sync	test-tag-from-devtools	<button>Sync</button>	
Periodic sync	test-tag-from-devtools	<button>Periodic sync</button>	

[Practical_6A_B](#) [Home](#) [Privacy](#)

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page Student Record

Roll No : 101

Name : Rahul